# SaSL-NLP Project (Milestone 1)

Semi-Supervised Learning for Text Classification

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March 5, 2026

# 1. Introduction to SSL in NLP

- **Context:** SOTA Deep Learning models (BERT, RoBERTa, XLNet) require massive labeled datasets.
- **Problem:** Manual labeling by experts is extremely expensive and time-consuming.
- **SSL Solution:** Leveraging unlabeled data to improve model performance and decision boundaries in classification.
- **Goal:** Reducing reliance on labeled data by learning from the vast amount of unlabeled text.

**Milestone 1**

Develop a pipeline that achieves high accuracy in low-resource regimes (only 1/5/10% of labels per class).

# 2. Possible Approaches in NLP

- **Pre-training & Fine-tuning:** Transfer learning from massive general-purpose corpora.
- **Pseudo-Labeling:** Generating "hard" labels from high-confidence model predictions.
- **TMix (Hidden Space Interpolation):**

$$\tilde{h} = \lambda h + (1 - \lambda)h', \quad \lambda \sim Beta(\alpha, \alpha)$$

- **Consistency Regularization:** Forcing stable predictions under perturbations.

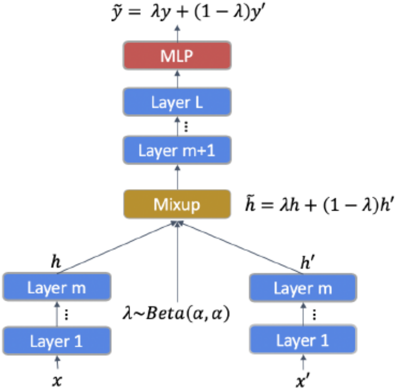


Figure: TMix: hidden-space interpolation between two text samples

# 3. Benchmark Datasets & Low-Resource Setup

| Dataset | Classes | Train | Test | Unlabeled Source | Task |
|---|---|---|---|---|---|
| AG News | 4 | 120,000 | 7,600 | Train split (minus labeled) | News classification |
| DBPedia 14 | 14 | 560,000 | 70,000 | Train split (minus labeled) | Ontology classification |
| IMDB | 2 | 25,000 | 25,000 | Unsupervised split: 50,000 | Sentiment analysis |
| Yahoo Topics | 10 | 1,400,000 | 60,000 | Train split (minus labeled) | Topic classification |

Table: Datasets with sample counts as reported on Hugging Face dataset viewers

- **Low-Resource Scenarios:** Only **1/5/10% labeled samples per class**; the remaining training examples are treated as unlabeled.

# 4. Performance and NLP Results

- **Data efficiency:** MixText performs strongly in low-label regimes by leveraging abundant unlabeled training data.
- **Generalization:** Consistency objectives reduce overfitting when labeled data is limited.
- **Smoother representations:** TMix encourages interpolation in hidden space, improving learned decision boundaries.

**Key highlight (from the paper)**

AG News (10 labels/class): **69.5%** (BERT) $\rightarrow$ **88.4%** (MixText)

# 5. Ablation Study & Architecture

- **Unlabeled Data:** performance drop observed when removed.
- **TMix:** Enhances representation diversity and smooth decision boundaries for different classes.
- **Sharpening:** Encourages confident and low-entropy pseudo-label predictions.
- **Augmentation:** Improves stability of the consistency regularization objective.

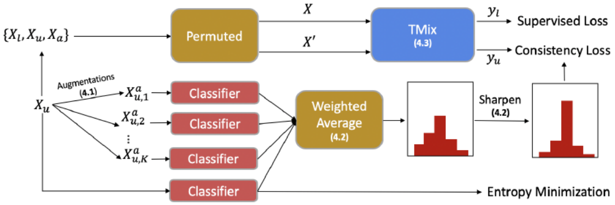


Figure: Overall MixText architecture with sharpening, TMix interpolation, and consistency regularization

# 6. Personal Contributions & Metrics

## Framework Development

- **Adaptive SSL Framework:** Modular pipeline for progressive unlabeled data usage.
- **Robust Pseudo-labeling:** Adaptive confidence thresholds and entropy-based filtering.
- **Semantic Consistency:** Constraint between original and back-translated text.

**Evaluation Metrics:**

- **Reliability:** Expected Calibration Error (ECE) and reliability diagrams.
- **Consistency:** Pairwise KL divergence and Top-1 agreement across augmentations.
- **Visualization:** t-SNE/UMAP embedding analysis for class separation.

# 7. Milestones Overview

**Milestone 1 (Done / Ongoing)**

Low-resource SSL baseline: MixText setup + training protocol + evaluation metrics.

**Milestone 2**

Run MixText experiments and report results (mean $\pm$ std) across label budgets.

**Milestone 3**

Implement SeqVAT and perform fair comparison: MixText vs SeqVAT under identical splits.

**Milestone 4**

Analyze similarities/differences: what helps both methods, what fails, and what is not "just SSL for NLP".

# 8. Milestone 2: Current Method (MixText) — Results

- Train MixText under **1/5/10% labels per class** on each dataset.
- Report: **Accuracy, Macro-F1, Precision, Recall, ECE**, stability across random seeds.
- Provide: 1 plot per dataset (*accuracy vs labeled budget*).

## Results Table Template (with future results)

| **Dataset** | **Labels/Class** | **Acc** | **Macro-F1** | **Precision** | **Recall** |
|---|---|---|---|---|---|
| AG News | 4 | TBD | TBD | TBD | TBD |
| DBPedia 14 | 14 | TBD | TBD | TBD | TBD |
| IMDB | 2 | TBD | TBD | TBD | TBD |
| Yahoo Topics | 10 | TBD | TBD | TBD | TBD |

# 9. Milestone 3: SeqVAT vs MixText — Comparison

- Implement **SeqVAT** (sequence-level VAT) with the same backbone and preprocessing.
- Keep experiments fair: same labeled budgets, same unlabeled pool, same evaluation.
- Output: **MixText vs SeqVAT** results + significance (mean $\pm$ std over seeds).

## Comparison Table (with feature results)

| **Dataset** | **Labels/Class** | **MixText (Acc/F1)** | **SeqVAT (Acc/F1)** | $\Delta$ |
|---|---|---|---|---|
| AG News | 4 | TBD | TBD | TBD |
| DBPedia 14 | 14 | TBD | TBD | TBD |
| IMDB | 2 | TBD | TBD | TBD |
| Yahoo Topics | 10 | TBD | TBD | TBD |

# 10. Milestone 4: Findings — Similarities and What Matters

## What helps both MixText and SeqVAT

- Strong unlabeled usage + stable training (seed robustness).
- Confidence control: calibration-aware thresholds / sharpening.
- Regularization that improves **smoothness** of predictions under perturbations.

## Key differences

- **MixText:** hidden-space interpolation to create diverse mixed representations.
- **SeqVAT:** adversarial perturbations that enforce local smoothness in sequence representations.